

Ocean Gyre Circulation

Sverdrup and Munk Solutions

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Introduction

Ocean circulation gyres are consequence of the large scale atmospheric circulation.

- 1 Heating air at the equator $\rightarrow$ Hadley cells.
- 2 Coriolis effect deflects the air movement $\rightarrow$ Trade winds and westerlies
- 3 Enforce wind stress on the ocean surface $\rightarrow$ 90° Ekman transport.
- 4 Since f increases with latitude, net southward Ekman transport in the North Atlantic.
- 5 Western boundary current needed to close the circulation.



Figure: A map of the North Atlantic Ocean.

Introduction, Sverdrup

Harald Ulrik Sverdrup (1888-1957) formulated the net transport in the ocean interior as:

$$-\beta \frac{\partial \Psi}{\partial x} = \frac{1}{\rho} \nabla \times \vec{\tau} \quad (1)$$

which gives the Sverdrup solution:

$$\begin{aligned} \int_x^{x_{\text{East}}} \frac{\partial \Psi}{\partial x} dx &= \int_x^{x_{\text{East}}} v H dx = \cancel{\Psi(x_{\text{East}}, y)} - \Psi(x, y) \\ \Rightarrow \Psi(x, y) &= -\frac{1}{\rho \beta} \int_x^{x_{\text{East}}} (\nabla \times \vec{\tau}) \end{aligned}$$

Introduction, Munk

Walter Heinrich Munk (1917-2019) extended the Sverdrup model by including meridional harmonic Newtonian viscosity (diffusion of momentum):

$$\boxed{A \nabla^4 \Psi - \beta \frac{\partial \Psi}{\partial x} = \frac{1}{\rho} \nabla \times \vec{\tau}} \quad (2)$$

which gives the Munk solution:

$$\Psi(x, y) = \Psi_{\text{Sverdrup}}(x, y) \left(1 - e^{-\frac{x - x_{\text{West}}}{2\delta_m}} \left[\cos\left(\sqrt{3} \frac{x - x_{\text{West}}}{2\delta_m}\right) + \frac{1}{\sqrt{3}} \sin\left(\sqrt{3} \frac{x - x_{\text{West}}}{2\delta_m}\right) \right] \right)$$

$$\text{where } \delta_M \equiv \left(\frac{A}{\beta} \right)^{\frac{1}{3}}$$

Stress set as:

$$\tau = \rho_{\text{air}} C_D U_{10}^2 \sin \left(\frac{\pi y}{L_y} \right). \quad (3)$$

The Sverdrup model:

- For the Atlantic Ocean: $L_x = L_y = 7000 \text{ km}$
- For the Pacific Ocean: $L_x = 14\,000 \text{ km}$ and $L_y = 7000 \text{ km}$.
- $\Delta x = \Delta y = 35 \text{ km}$
- $U_{10} = [4 \text{ m s}^{-1}, 5 \text{ m s}^{-1}, 6 \text{ m s}^{-1}]$

The Munk model:

- $L_x, L_y, \Delta x$ and Δy same as before.
- $U_{10} = 5 \text{ m s}^{-1}$
- $A = [3 \times 10^4 \text{ m}^2 \text{ s}^{-1}, 1 \times 10^5 \text{ m}^2 \text{ s}^{-1}, 5 \times 10^5 \text{ m}^2 \text{ s}^{-1}]$.
- $\Rightarrow \frac{\delta M}{\Delta x} \approx [3.3, 4.9, 8.4]$

Result, The Sverdrup solution

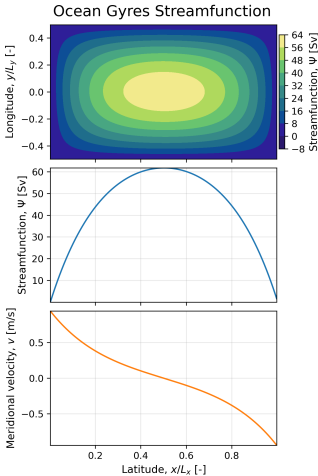


Figure: Streamfunction for the no-drag f -plane configuration. The circulation remains symmetric with no β -effect and without western intensification.

Result, The Sverdrup solution

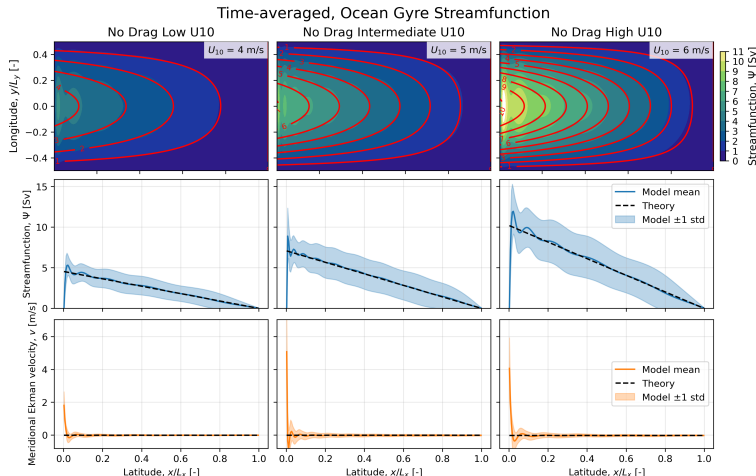


Figure: Streamfunction for the no-drag β -plane configuration. The red lines are the theoretical Sverdrup solution.

Result, The Sverdrup solution

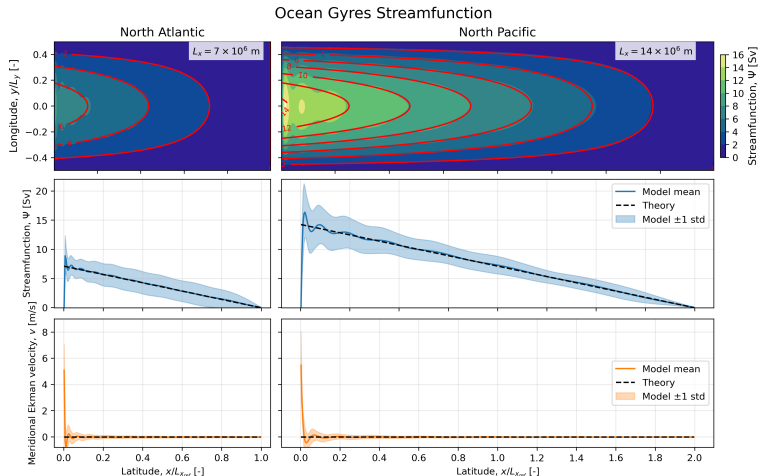


Figure: Streamfunction for the no-drag β -plane configuration with varying domain size. The red lines are the theoretical Sverdrup solution.

Result, The Munk solution

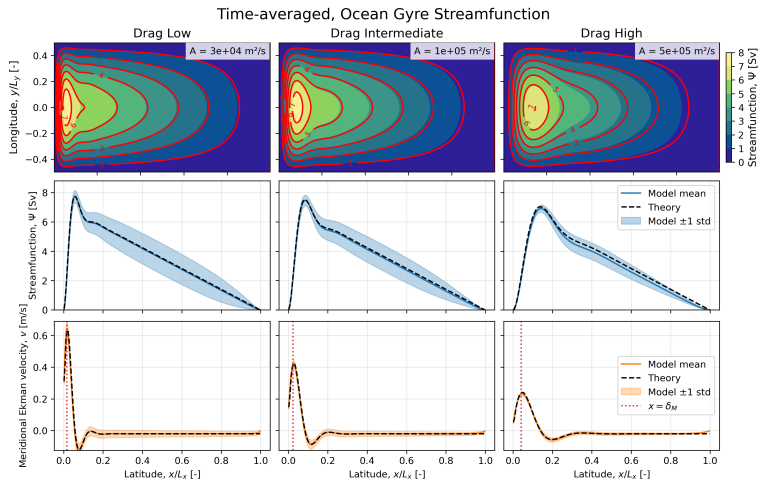


Figure: Streamfunction for the Laplace-viscous β -plane configuration. The red lines are the theoretical Munk solution.

Result, The Munk solution

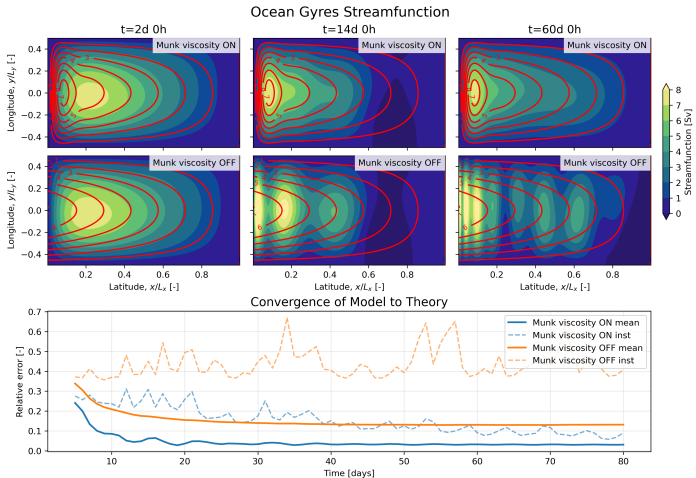


Figure: Streamfunction for the Laplace-viscous β -plane configuration. The red lines are the theoretical Sverdrup and Munk solutions respectively.

Discussion

Both models show western boundary currents due to mass conservation, but only Munk theory shows a western boundary current. Why?:

- The Sverdrup theory is linear, only one B.C. $\Psi(x_{\text{East}}) = 0$.
- The Munk theory is non-linear, four B.C.s: $\Psi(x_{\text{East}}) = \Psi(x_{\text{West}}) = v(x_{\text{East}}) = v(x_{\text{West}}) = 0$.
- Frictional force is opposite sign to the motion itself, needed to balance Sverdrup flow.
- The Munk viscosity creates a boundary layer where the frictional force can take care of mass conservation.

$$A\nabla^4\Psi - \beta\frac{\partial\Psi}{\partial x} = \frac{1}{\rho}\nabla \times \vec{\tau}$$

Conclusion

Important conclusions:

- Both numerical models showed western intensification, explained by Rossby wave propagation.
- Only Munk theory showed a western boundary current, explained by frictional layer, although both numerical models showed return flow close to the western boundary.
- A larger domain gave same interior meridional transport, which resulted in a stronger western boundary current.